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Identification of orcein and selected natural dyes in 14th and 15th century liturgical paraments with high-performance liquid chromatography coupled to the electrospray ionization tandem mass spectrometry (HPLC-ESI/MS/MS)

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Abstract

In this work, a novel method for identification of orecin derivatives from orchil is presented. It was developed, similarly to other methods used to determine natural dyes, with the use of liquid chromatography coupled to tandem mass spectrometry (LC/MS/MS). Apart from presenting a novel method for orecin derivatives determination, this work also aims to review currently existing method for identification of blue (indigo), red (anthraquinones and brazilwood), and yellow (flavonoids) dyes. In described research, selected reaction monitoring (SRM) conditions were optimized for a number of natural colorants, including the characteristic components of blue (indigo), red (anthraquinones, orchil and brazilwood) and yellow (flavonoids) dyes. For the first time, the major colorants of orchil (orcein derivatives) were analyzed with high performance liquid chromatography (HPLC) coupled to the triple quadrupole mass analyzer. This, made the developed analytical method very sensitive and selective with LOD's values in the range of 0.002 and 0.09 µg/ml (when pure standards were available). HPLC coupled to the time-of-flight mass analyzer (TOF/MS) was used to define elemental composition of the orcein derivatives at given retention times. The analysis of natural indigo allowed to conclude that indigo prepared using *Isatis tinctoria* and *Indigofera tinctoria* contained similar amounts of indirubin. However, the isatin content was significantly higher for *indigofera tinctoria* than for *istais tinctoria*. Before analyzing historical samples, analysis of the natural indigo were carried out in order to evaluate if isatis to indigotin ratio can be used to elucidate the plant source of indigo detected in historical samples. Commercial *Indigofera* indigo powder showed significantly higher value of this ratio (above 0.5) than *Isatis tinctoria* for which this ratio took values below 0.2. This proportion was used for the identification of blue dye's source. Therefore, the difference in isatin content may be used to identify indigo plant source. This approach requires careful and accurate analysis because isatin is not only a component of natural indigo but also its decomposition product. In this work, the developed analytical protocol was used to successfully identify natural dyes: indigo, brazilwood and orchil in medieval silk textiles (14th and 15th century) from collection of liturgical vestments from Saint Mary's Church in Gdańsk (Poland), currently stored in National Museum in Gdańsk.

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1. Introduction

Indigo is one of the oldest dyes known to mankind [1-3]. The oldest indigo-dyed objects are dated back around 3500 (salt mine in Hallstatt) [4] or 4000 years ago (textiles discovered in Tutankhamen's tomb c. 1350 BC) [2-5].However, the earliest transcribed procedures of dying with indigo were found in the *Papyrus Graecus Holmiensis* from 3rd century AC [4]. In Europe for centuries, the main plant source of indigo dye was *Isatis tinctoria.* It changed with the extensive overseas exploration in 16th-17th century. *Isatis tinctoria* was replaced with cheaper *Indigofera*, originating from Indie. The competition between these two indigo plant sources in Europe ended in late 1890s when synthetic indigotin was introduced [1].

Natural indigo is produced from plant precursors: indican (mainly found in *Indigofera tinctoria*) and isatan B (a major indigo precursor in I*satis tinctoria*) [1-3 ,6] . The process of obtaining blue dye from both plants, typically is twofold. First, dry leaves are subjected to a fermentation process, and as the result indoxyl is formed [4]. Second, indoxyl under the influence of mixture containing stale urine, lime or ash lye in alkaline conditions transforms to colorless leuco-indigotin [3, 4]. Obtained in such a process plant originating dye is later used for colorization textile fabrics. The textile fibre, however, do not change its color as the result of the leuco-indigotin bath but as the result of exposing soaking wet fabrics to air; colorless leuco-indigotin transforms to blue indigotin [3]. Although, indigo synthesis begins from different precursors, the dye composition, prepared from both plants, is very similar [2, 6]. For this reason, accurate identification of indigo's plant source is often difficult and elusive [4.6-8].

In the past, high content of indirubin was considered to be characteristic for indigo produced from *Indigofera tinctoria* [8, 9]. However, according to recent results found in literature this assumption might be seen as dubious. The indirubin content in indigo dye prepared either from *Isatis tinctoria* or *Indigofera tinctoria* are indistinguishable [2, 6-8, 10].

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Alike doubts arise about other characteristic compound of indigo dyes - isatin. Mantzouris et al. [11] reported that analysis of the commercial indigo, produced from *Indigofera tinctoria*, revealed the presence of significant amount of isatin. However, similar analysis carried out on Kremer indigo, often used as reference material, did not yield similar results [4, 7, 12, 13].

Identification of the raw material, used for natural dye preparation, based on the relative amounts of its main component (rather than relying on a specific marker) is not a new concept. For example, coccid dyes are tentatively identified based on the relative ratios of kermesic, flavokermesic and carminic acids [2, 14, 15]. The stability of natural dyes' constituents needs to be evaluated before calculating their relative concentrations [16] because for example isatin is simultaneously indigotin's decomposition product and its constituent [7, 12. 17] In this work, the relative amounts of indigotin, indirubin and isatin were investigated to identify the source of indigo in historical textiles.

Popular and stable natural dyes (such as indigo) were often used together with lesser dyes, including orchil and brazilwood, to secure the desired shade of blue or red. Purple dyes could have been obtained either as a result of mixing orchil with indigo [4, 18, 19] or as a result of adding brazilwood to indigo [20-22]. Red dye received from expensive kermes (*Kermes vermilio*), was sometimes imitated by a mixture of madder and brazilwood [23, 24]. These dyeing methods are well documented in the literature and are discussed in section 3.3 (case study).

Numerous studies have shown that LC/MS is the most reliable method for identifying natural dyes found in the micro-samples from the objects of historical value [8, 14, 25]. However, LC/MS protocols have limited number of identified characteristic compounds for dyes. Therefore they allow to only define a set number of dyes. Exploration for new potential markers might enable to better recognize the source of the dye, and what is more to better

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understand the origins of the fabrics. Detailed information about dyes used in objects of historical value, potentially, allow to understand not only the origins of the textile but also when they were made and for whom.

There were two major objectives of the experimental work presented in this paper. First, it was evaluated if the amount of isatin in the sample can be used to identify the plant-source of indigo. In 14th-15th centuries, *Isatis tinctoria* was the most popular source of indigo in indigo-containing textiles in Europe. However, the beginning of transcontinental trade exchange brought to the European markets *Indigofera* orginated indigo dye from Asia. The imported Asian *Indigofera originated dye* was more expensive but was also considered as a higher quality dye than the one obtained from *Isatis tinctoria* [26, 27].

Second, LC/MS/MS with electrospray ionization (ESI) method for identification of most popular and lesser natural dyes was developed. Brazilein, brazilin, hematoxylin and hematein - the main components of redwood dyes were analyzed with direct infusion ESI/MS/MS [28]. LC/MS analysis conditions for hematoxylin and hematein were described by Pawlak et al.[29]. Gulmini et al. [30] described analysis of brazilein with HPLC-DAD-MS. Here, a procedure for simultaneous quantification of four neoflavonoids: brazilein, brazilin, hamatoxylin and hematein in historical textiles using the LC/MS/MS is presented. Up-to-date, only one paper, published by Rosi et al., [31] describes orchil analysis with LC-TOF/MS; however the results are ambiguous (for more details see section 3.2.1). This paper is therefore a first investigation describing analysis of all of the major components of orchil with LC/ESI-MS/MS instrument operating in the SRM mode. Identification of indigo, orchil and brazilwood in 14th and 15th century liturgical paraments from Saint Mary Church in Gdańsk (Poland) underlines that this protocol can be used to successfully identify natural dyes in historical textiles.

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2. Experimental

1. Materials and reagents

Indigotin (≥ 95%), indirubin (≥ 98 %), isatin (≥ 98 %), LC-MS eluent additive - grade formic acid (≥ 99 %), hematoxylin (≥ 95%), hematein (≥ 95%), orchil (Natural red 28), purpurin (≥ 90%), alizarin (≥ 97%), emodin (≥ 90%) and anthragallol (≥ 90%) were purchased from Sigma – Aldrich (Schnelldorf, Germany). Quercetin (≥ 99%), apigenin (≥ 99%), luteolin (≥ 99%), rhamnetin (≥ 99%), morin (≥ 99%), kaempferol (≥ 99%), rutin (≥ 99%), genistein (≥ 99%) and fisetin (≥ 99%) were purchased from Extrasynthese (Lyon, France). Acetonitrile and methanol (hyper-grade for LC-MS) as well as copper foil were supplied by Merck (Warszawa, Poland). Dimethyl sulfoxide (DMSO, ≥ 99.7%), sulfuric acid (≥ 95%), aqueous solution of hydrochloric acid (HCl, A.R, 35-38%) were obtained from Avantor (Gliwice, Poland). Natural indigo prepared from *Indigofera tinctoria*, I*satis tinctoria* and brazilwood dye powder were purchased from Kremer (Kraków, Poland). Nitrogen (≥ 99.5%) was supplied by Multax (Stare Babice, Poland). Deionized water (18 MΩ×cm) was prepared using Direct - Q3 Ultrapure Water System (Millipore). PTFE membrane syringe filters (pore size 0.2 µm) were obtained from Sigma – Aldrich (Schnelldorf, Germany).

2. Sample preparation

Initially, 400 µl of 37% aqueous HCl/methanol/water (2:1:1, v/v) was added to the sample (2 – 13 mg). Afterwards, sample was heated for 10 min in dry heating block at 100 °C. After 10 minutes of heating the solvent was evaporated under a gentle stream of nitrogen at 60 °C and 600 µl of methanol/water solution (2:1, v/v) was added. Dyes soluble in water/methanol mixture were extracted for 10 min by sonification. The solution containing mordant as well as direct dyes were separated from the residue by centrifugation at 14000 RPM for 15 min and

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filtered through 0.2 µm PTFE syringe filter before the chromatographic analysis. 10 µl of the solution containing vat dyes was subjected to the chromatograph.

The extraction residue was again dried under a stream of nitrogen at 60 °C and 600 µl of DMSO/ACN solution (1:1, v/v) was added to the sample. Afterwards, vat dyes were extracted by sonification for 30 min at 60 °C. The sample was centrifuged for 10 min at 14000 RPM and the solution was filtered through PTFE syringe filter (pore size 0.2 µm). 5 µl of the solution containing vat dyes was subjected to the chromatographic analysis.

3. Apparatus

Analysis were performed with LC20 liquid chromatograph (Shimadzu) coupled to QTRAP 3200 mass spectrometer (Applied Biosystem/MDS SCIEX). Analytes were separated with Luna (Phenomenex) C18 column (100 mm × 2.1mm, 3 µm, 100 Å), the column over temperature was set to 30°C and the analytical column was equipped with 2mm ID C18 pre-column. Aqueous solution of formic acid (pH 2.8) and acetonitrile (ACN) were used respectively as mobile phase components A and B. The mobile phase was delivered at a flow rate of 0.2 ml/min in gradient mode: 0–2 min 5% B, 2–30 min linear gradient to 60% B, 30–35 min linear gradient to 100% B, 35-40 min 100% B, 40-41 min linear gradient to 5% B. Afterwards, the column was re-equilibrated at 5% B for 14 min and the analysis was completed in 55 min. Mass spectrometer was operating in the selected reaction monitoring (SRM) mode; mass spectrum in the total ion current (TIC) mode was also acquired in the range of 50 – 650 m/z. Optimized electrospray ionization (ESI) conditions were: source temperature 450 °C, nitrogen used as curtain (3×10^5 Pa), auxiliary gas (3×10^5 Pa) and collision gas. Mass spectra (TIC and SRM mode) were acquired in the negative and positive ionization modes and the capillary voltage was -4.5 kV and +5.5 kV, respectively. MS/MS

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conditions for each compound were optimized in the infusion mode as described in section 3.1. Analyst 1.4.2. software was used to acquire and process the data.

High-resolution mass spectrometry (HR-MS) analysis were carried out with Q-TOF 6540 Ultra High Definition mass spectrometer (Agilent) equipped with the dual AJC ESI ion source. TOF mass spectrometer was connected to the 1290 Infinity liquid chromatograph (Agilent) and the HPLC analysis conditions (including the column and mobile phase composition) were the same as for the low-resolution experiments carried out using triple quadrupole mass spectrometer. Q-TOF/MS was operating in TIC mode in the mass range 100-1200 m/z and the ESI conditions were as follows: drying gas (N_2) flow: 8 L/min, nebulizing and sheath gas (N_2) temperature: 350 °C, nebulizing gas pressure: 2.4×10^5 pa and the ESI voltage was 3.5 kV. Mass calibration standard was delivered into the ion source via second ESI capillary in the dual AJC ESI ion source. Mass Hunter VB05.01 software was used for data acquisition and processing. For the elemental formulas assignment, detected ions were assumed to be singly charged, $[M+H]^+$ protonated pseudo-molecular ions containing C, H, O and N atoms. Elemental formulas with the smallest difference between measured and calculated m/z values are listed in Table 2 in section 3.1.2.

4. Indigo dyeing method

Indigo-dyed silk was prepared in the laboratory in order to evaluate the impact of the dyeing method on the dye composition. Unfortunately, the fresh plant materials for both *Isatis tinctoria* and *Indigofera tinctoria* were not available, thus a dyeing method utilizing commercial indigo powder was used. This procedure is known as Nencki dyeing method, in which indigotin reduction is carried out in copper kettle under acidic conditions [32]

Following Nencki dyeing method, first, two pastes were prepared by mixing 600 mg of commercial *Indigofera tincotoria* or *Isatis tinctoria* indigo powder with 2.5 ml of 98%

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sulfuric acid. Afterwards both mixtures were left alone for 24 hours. After 24 h two dyeing baths were prepared in two separate copper pots, each consisting of 400 ml of water and 8 to 12 drops of one of the acidic pastes. When the initially deep - blue solution was decolorized, silk was placed inside the pots and the baths were heated to 90°C; this temperature was maintained for 1 h. After 1 hour dyed silk was removed from the dyeing baths, rinsed in 100 ml of distilled water and left to dry overnight. This procedure was repeated several times in order to obtain deeper shade of blue.

5. Historical samples

As the part of the detailed analysis of 192 medieval liturgical vestments currently stored in the National Museum in Gdansk, the identification of selected natural dyes present in these textiles was carried out. This paper describes a portion of these analysis, focused on studying the presence of indigo, orchil and brazilwood.

The inventory from the year 1552 indicates that the endowment of the church St. Mary Church was impressive (46 altars and chapels fully equipped with liturgical paraments, including 229 chasubles). In the year 1572 the St. Mary's Church passed under the Lutheran administration. The liturgical vestments were then gradually withdrawn from use. Stored in cases, closed in altar vaults, they have been forgotten over time. They were gradually discovered during the renovation work, carried out in the church after the year 1820. The largest part of the collection was found by August Hinz in the years 1861-1864. In the 19th century the vestments have been partially dispersed due to the sale organized in order to raise funds for the church renovation. In the years 1930 - 1933 Walter Mannowsky, at the time the director of the Museum of the City of Gdansk, saw the value, great historical and artistic importance of this collection, and conducted an inventory of the remaining vestments. His work, completed by 1937, developed into a five-volume collection catalog which included

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541 vestments. In 1937 the whole collection, as a deposit, was handed over to the Museum of the City of Gdansk (Danziger Stadtmuseum). During the Second World War the collection was transported to Germany. In 1961, only a part of it returned to Gdansk. Out of 541 objects only 192 were returned to Poland. Other objects from the collection of pre-war Stadtmuseum in Gdańsk are now in Lübeck in St. Annen Museum and the German National Museum in Nürnberg.

6. Statistical analysis

Cluster analysis was carried out with Statistica software (Statsoft, Poland). Data vector for each sample contained relative indigotin, indirubin and isatin concentrations (%, w/w) – see section 3.2. Before the tree diagrams were constructed, Euclidean distances were calculated and the samples were assigned into the individual clusters using Ward's clustering method.

3. Results and discussion

3.1. Method development

Initially, standard solutions were injected directly into the triple quadrupole mass spectrometer ion source using a Harvard Apparatus pump at a flow rate of 10 μL/min. The following MS/MS parameters were optimized for the each Q1/Q3 transition (SRM): entrance potential (EP), declustering potential (DP), collision cell entrance potential (CEP) for the deprotonated pseudomolecular ions, the collision energy (CE) and collision cell exit potential (CXP); optimal values of these ion lenses voltages are listed in Table S1.

For each analyte the most intense Q1/Q3 transition was selected in order to achieve the highest sensitivity. The characteristic SRM transition together with the retention time of the analyte allowed for unambiguous identification of the given dye in the sample taken from the historical textile. After optimizing the qualitative parameters, quantitative analysis was carried

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out. Standard solutions of indigotin, indirubin and isatin were prepared in DMSO/ACN (1:1, v/v). Other dyes listed in Table 1 were dissolved in methanol/water (2:1, v/v). The standards' concentrations for vat, direct and mordant dyes were 0.0005 – 0.02 mg/ml. Standards were injected into the liquid chromatograph and analyzed under the experimental conditions described in section 2.3. All analysis were carried out using triple-quadrupole instrument unless stated otherwise. Parameters of the developed quantitative analysis method are listed in Table 1.

As listed in Table 1 for every analyte (when pure standard or purified dye was available) squared linear regression coefficient (R^2) was > 0.99 when the integrated peak area and analyte concentration expressed in mg/ml were used as dependent and independent variables, respectively. Limit of detection (LOD) and limit of quantification (LOQ) were obtained by calculating the concentration corresponding to the average background signal plus three (LOD) and ten (LOQ) standard deviation values of the background noise [33]. Chromatograms of the standard solution are shown in Fig. S1 (flavonoids), S2 (anthraquinones), S3 (neoflavonoids), S4 (orchil), and S5 (indigo, indirubin and isatin). Only the extracted ion chromatograms (EIC) for the most intense SRM, used for the quantitative analysis, are shown in Fig. S1-S5.

For the analytes listed in Table 1 method's sensitivity (expressed as LOD and LOQ values) was sufficient for natural dyes identification in micro-samples from 14th and 15th historical textiles, as described in section 3.3. Standard solutions were analyzed at the beginning and at the end of the sample batch to monitor the mass spectrometer response factors for each analyte. Reported uncertainty is two values of the standard deviation unless stated otherwise. The instrument was recalibrated regularly to ensure that the results were reproducible. Initially, analysis of natural indigo was carried (section 3.2) before historical samples were analyzed (section 3.3).

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3.1.1. Indigo

In this work, indigotin, indirubin and isatin were selected as markers of indigo presence in the sample. These compounds are well-known indigo chromophores and it is reasonable to assume that they compose a majority of the natural indigo colorants [4, 6, 7, 34]. In the past, both atmospheric pressure chemical ionization (APCI) [13, 35] and ESI [34, 36] were used for indigo identification in the historical textiles. In this work, indigotin, indirubin and isatin were ionized with ESI, since all of the compounds listed in Table 1 can be ionized using this method [9, 13, 28, 30, 31, 37, 38].

Previously, ESI was used to detect indigo constituents in the positive [7, 29, 34, 36, 39, 40] and in the negative [11, 13, 29,40] ionization modes. For indigotin and indirubin, the ionization efficiency was significantly higher in the positive ionization mode; this could be explained by the use of acidic eluent as pointed out by Lech et al. [40]. Some problems were encountered for isatin: high background noise was detected for the SRM 148/120 (data not shown) resulting in higher LOD and LOQ values for isatin in the positive ionization mode [36]. Therefore, MS/MS conditions were optimized for indigotin and indirubin in the positive ionization mode and isatin was detected in the negative ionization mode. The most intense Q1/Q3 transitions listed in Table 1 were consistent with the literature data [7, 13, 36, 39]. Values of LOD's and LOQ's for indigo components were also comparable with the numbers reported previously [7, 36].

3.1.2. Orchil

Unfortunately, pure standards were not available for orcein derivatives (for brazilin and brazilein pure standards are also unavailable, see section 3.1.3) and dye powder was used instead for qualitative analysis of these compounds. Such approach is less accurate than the calibration carried out with high purity standards but still provides an approximation of a

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given dye content in the sample. The actual LOD and LOQ values for the individual orcein derivatives are probably lower than the ones listed in Table 1 [37]. Structures of the known orcein derivatives are listed in Table 2 [2, 41-43].

LC-HR-MS analysis of orchil (see section 2.3) were carried out in addition to the MS/MS analysis, since currently available literature data regarding chromatographic data and accurate mass data of the compounds listed in Table 2 is ambiguous [31]. EIC chromatograms, acquired using LC-ESI/TOF/MS instrument, for α, β and γ orcein derivatives are shown in Fig. 1.

As shown in Fig. 1 all of the major orcein components were detected in the synthetic dye [41]. As reported previously [2, 41-43] only α-amino orcein and α–hydroxy orcein have distinctive elemental compositions. β and γ derivatives consisted of two structural isomers (Table 2), thus distinguishing these molecules was not possible. α-amino orcein and α–hydroxy orcein were the major synthetic orcein components estimated from the integrated peak areas (TIC). Rosi et al. [31] reported that the measured m/z of α-amino orcein $[M+H]^+$ ion was 362.8916 Da. Elemental formula of α-amino orcein is $C_{21}H_{18}N_2O_4$ and the monoisotopic mass of this compound $[M+H]^+$ ion is 363.1339 Da – a number significantly different from the value reported by Rosi et al. [31]. Therefore, it is important to underline that data obtained in this work is reliable and consistent with the expected elemental composition of the orcein derivatives. Moreover, identification of orcein derivatives in the SRM mode allows for unambiguous identification of this dye in historical samples, which would not be possible if likewise previous studies diode array detector was used (DAD) [44].

As expected, for all compounds listed in Table 2, analysis sensitivity was higher in the positive ionization mode than in negative ionization mode. It is well known that amines can usually be detected with ESI operating in the positive ionization mode [45]. Also, under

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acidic conditions used here (section 2.3.) it is reasonable to assume that the analytes listed in Table 2 are already ionized in the liquid-phase, which leads to the enhanced ionization efficiency in the positive ionization mode [45]. Therefore, MS/MS conditions for orcein derivatives were optimized for the $[M+H]^+$ pseudomolecular ions. Significant quantities of minor orcein components, including orsellic acid, lecanoric acid and erythrin [41] were not detected in the synthetic dye.

3.1.3. Neoflavonoids

Previously, identification of hematein, hematoxylin, brazilin and brazilein in historical textiles was carried out with HPLC-UV/Vis-based methods and the results were often inconclusive [46, 47]. Lech et al. [40] reported that hematein and hematoxylin ionization in negative ESI under acidic conditions was problematic. However, other researchers described successful analysis of these compounds with LC/MS utilizing electrospray ion source [29, 30, 37, 48]. Fragmentation mechanism of $[M-H]^-$ pseudomolecular ions for hematein, hematoxylin, brazilin and brazilein was studied by Hulme et al. [28]. Analysis of these four neoflavonoids were carried out in the SRM mode thereby allowing for unambiguous identification of the redwood dyes under study.

In previous studies sensitivity analysis for the redwood dyes were never reported. In this paper, additional experiments were carried out to compare the three detection methods (UV/VIS, MS in SIM mode, and MS in SRM mode). Data presented in Table S2 indicate that sensitivity in the SRM mode was one to two orders of magnitude lower than in MS (SIM) and UV-Vis. This results are consistent with the results obtained for other dyes [37, 49].

3.1.4. Flavonoids and anthraquinones

Antraquinone and flavonoid dyes recently were identified in historical textiles with the use of LC/MS [13, 34, 37, 50-54]. SRM pairs for the major components of red and yellow

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dyes of plant origin were consistent with the literature data[13, 34, 53, 54]. Antraquinones studied in this work: purpurin, alizarin, anthragallol and emodin are found in the dyes prepared from madder (*Rubia*) [14, 34, 51, 52] or lady's bedstraw (*Galium verum L.* and *Galium mollugo L.*) [14, 52]. Flavonoids allowed for identification of the most frequently encountered yellow dyes, such as young and old fustic (*Cotinus coggygria Scop.* and *Maclura tinctoria*, respectively) [37] and Weld (*Reseda luteola L.*) [34] and Persian berries (*Rhamnus*) [38].

3.2.Natural indigo analysis

Analysis of natural indigo was carried out to evaluate if there were any significant differences in the composition of indigo powder prepared from *Isatis tinctoria* and *Indigofera tinctoria.* Additionally it was inquired if this data could be used to identify the plant-source of indigo in historical textiles. Natural indigo powder and dyed silk samples, prepared in the laboratory, were extracted with DMSO/ACN mixture (1:1, v/v).

In the past, a number of different solvents were used for indigo extraction from historical samples. DMSO [11, 13, 29, 34, 55] is the most popular indigo extraction solvent followed by DMF [11, 51, 56] and pyridine [51,56]. In this work, ultrasound-assisted extraction of indigo at 60 °C was carried out. Indigo extraction at elevated temperatures [7,11,14,55] yielded satisfactory results for indigotin, indirubin and isatin [11]. Results of the preliminary studies revealed that under the experimental conditions (described in section 2.3) the use of pure DMSO resulted in peak tailing in chromatography of the indigo components under study. Replacing extraction solvent with DMSO/ACN mixture (1:1, v/v) eliminated this problem [7].

As shown in Fig. 2 the three indigo components under study were detected in the natural dye prepared from *Isatis tinctoria*. Integrated SRM peak areas were used to calculate the concentration of the analytes in the natural indigo powder.

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As shown in Fig. 3A and Fig. 3B natural indigo composition varied, depending on the dye source. As already discussed in the introduction, higher indirubin content in the sample was characteristic for the indigo prepared from *Indigofera tinctoria* plant material [51]. However, currently available data indicates that higher indirubin concentration is not characteristic for indigo prepared from either *Isatis tinctoria* [6, 7] or *Indigofera tinctoria* [9].

Results of the GC/MS study carried out by Degani et al. [6]showed that xylitol and gallic acid were present only in the commercial *Indigofera* indigo. Hartl et al.[4] reported that flavonoids were present only in the silk dyed with *Isatis tinctoria* – plant material was used for dyeing and the process was carried out according to the historical recipes. However, flavonoids, gallic acid and xylitol are not indigo-characteristic compounds and their presence in the sample can arise from a number of different reasons [4, 6, 8].

Abdel-Ghani et al. [10] studied the commercial Kremer natural indigo compositions in order to elucidate the plant source of blue dye found in the 18th century Coptic icons. The authors concluded that the indigo detected in these icons probably originated from *Indigofera* leaves. They based their claims on the presence of characteristic inorganic salts detected in both 18th-century indigo and Kremer-brand *Indigofera* indigo.

In this work it was investigated if the relative isatin content in the indigo dye could be used to elucidate the plant source of indigo in a very well preserved samples. As shown in Fig. 3 isatin content of the commercial dye powder depended on the indigo plant source, as opposed to indirubin. The isatin to indigotin ratio remains the same for both plant's extract solution and textiles coloured with it. Significant differences in this ratio between source plant allow to define whether *Isatin tinctoria* or *Indigofera tincotira* was used to prepare the dye. Isatin to indigotin ratio above 0.5 allows to predict *Idigofera tinctoria* as a dye source and values below 0.2 - *Isatis tictoria.* Although, isatin is a component of both plants on which

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indigo dye is based, it is also a product of indigotin degradation, hence its concentration might be an effect of decomposition not only the plant source [57]. However, objects examined in this paper came from 14th and 15th century textiles, which were stored in dark and dry conditions. Thus, it is allowed to assume that for these textiles indigo decomposition was very slow, perhaps even negligible therefore did not affect proposed plant source recognition method. When dyeing method described in section 2.4 was used, the dye composition was similar to the natural indigo powder (starting material). However small increase in isatin/indigotin and indirubin/indigotin concentrations ratios were observed.

It is difficult to discuss the mechanistic details of the redox processes taking place in the dyeing bath, since only the final dye composition was studied. Unfortunately, it was not possible to examine the intermediate stages of the reactions, as opposed to e.g. studies of indigo photodegradation in the solution [12, 34].

In order to evaluate if small changes in the dye composition allowed for the indigo plant source identification, statistical analysis was carried out. As shown in Fig. 4, plant source of indigo in silk samples, dyed in the laboratory, might be correctly identified, despite small changes in the dye composition during the reduction/oxidation reactions taking place in the dyeing bath.

Results presented in this section indicate that after dyeing the silk with natural indigo, dye composition extracted from the laboratory-prepared samples was only slightly different than in the indigo powder, initially used to prepare the dyeing bath. Small increase in the isatin/indigotin and indirubin/indigotin ratios were observed but the statistical analysis showed that the plant source of the dye can still be correctly identified. These conclusions are accurate for samples dyed with commercial-Kremer dyes using metallic copper/sulphuric acid as reducing agents during dyeing bath preparation.

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Since Kremer-brand indigo is frequently used as reference material in studies of natural dyes in historical samples [4, 7, 10, 12, 13] an important question arises: is the composition of commercial dyes similar to the indigo composition prepared a few centuries ago? Namely, does indigo prepared from fresh *Isatis tinctoria* leaves contain high amounts of isatin and is the isatin content enhanced by dyeing the textiles with fresh plant material? Results of comprehensive investigation carried out by Hartl et al. [4] provided some insights into the dye composition, prepared from the fresh plant-material. They determined isatin concentration in dye baths and silk dyed with the use of different extracts. In their work they used extracts of *Isatis tinctoria* fresh leaves, green and couched *Isatis tinctoria*, and *Isatis tinctoria* commercial pigment (Kremer's dye). Results showed that isatin concentration in all samples was low and did not vary in-between. Unfortunately, such a research can be found only for *Isatis tinctoria* and it is currently unclear whether or not indigo prepared from fresh plant material from *Indigofera tinctoria* contains high amounts of isatin. In this study, the comparison was made between samples acquired from Kremmer's commercial pigments originating from *Isatis tinctoria* and *Indigofera tinctoria*. Results showed that isatin concentration in *Indigofera tinctoria* was significantly higher than in *Isatis tinctoria.*

3.3.Case study – dye analysis of 14th and 15th century silk textiles from liturgical vestments

Historical samples, weighing 2-13 mg, were taken from the warp and weft yarns of 14th and 15th century silk textiles, preserved in liturgical vestments from collection of St. Mary's Church in Gdańsk (see section 2.5). They were prepared as described in section 2.2 and analyzed under the experimental conditions described in section 2.3.

Sample pretreatment included hydrolysis with 37% HCl/methanol/water (2:1:1, v/v) solution followed by extraction of mordant and direct dyes with methanol/water mixture.

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Such a procedure is a well-established and very efficient method for extracting wide variety of natural dyes [40, 52]. The major drawback of this method is partial hydrolysis of the less stable dyestuff components (such as flavonoid glycosides) [40, 58]. However, substituting HCl with organic acids in order to preserve more labile compounds was proven to significantly lower effectiveness of the extraction [14, 40].

Generally, medieval textiles under study were characterized by a well-preserved blue color. However, the degree of the direct and mordant dyes degradation (including orchil and redwood dyes) was significantly higher than in indigo. It is well known that the light fastness of indigo is very high as opposed to the lesser dyes studied in this work, namely orchil and brazilwood [2, 59]. Therefore, it was important to ensure the highest possible sensitivity of the analysis – for this reason samples were treated with HCl solution instead of organic acid. Extraction conditions' impact on the orchil recovery (with and without indigo presence) has not been studied.

Since indigo constituents are characterized by a very low solubility in water and alcohols [11, 51, 55], additional extraction with DMSO/ACN mixture was carried out (section 2.2) after the initial sample pretreatment with the HCl and extraction of direct and mordant dyes [9, 30, 49, 58]. The calibration curves (section 3.1) were used to evaluate the relative content of the natural dyes detected in the historical textile. Thus, it was possible to distinguish between the major and minor colorants.

Sample number 37 (sky blue/violet) was taken from the lampas (Italy, 2nd half of 14th century) preserved in the fragment of chasuble (inv. no. MNG/SD/241/TH) – Fig. 5A. As shown in Fig. 5C orcein derivatives were detected in this sample, thereby unambiguously confirming the presence of orchil [8, 41, 43]. This sample also had low value of isatin to indigotin ratio (below 0.1) which indicated woad (*Isatis tinctoria* L.) as source of blue dye.

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Orchil was identified only in sky blue main warp of the lampas. It formed the background composed of the sky blue, white, beige and gold patterns. The largest part of sky blue pattern is almost invisible on a very similar background. The identification of orchil indicated that the original background was violet or violet-blue but due to low light fastness of orchil the original color faded away relatively quickly [18]. As shown in Fig. 5B, the portion of the silk that was not exposed to light is still violet. The sky blue pattern is well visible, and can be easily distinguish from the background. Presented results revealed that the violet or gray shades of blue were obtained by using orchil, since no samples contained mixture of indigo or woad with red dyes alone (see Table 3). Results of LC/MS/MS analysis for the sample number 37 also confirmed the use of orchil in dyeing silk fabrics in Italy, as well as in Near East. Sample number 37 also contained trace amount of apigenin, as shown in Fig. 5C. The presence of this compound was most likely a result of sample contamination, thus the use of the yellow dye originating from plant such as weld (*Reseda luteola*) was tentatively excluded.

SRM chromatograms of the sample number 54 (main weft, scarlet color) taken from the orphrey of the chasuble dated to 14th or 15th century most likely originating from Italy or Spain are shown in Fig.6 – the photograph of this object is also included in Fig. 6. As shown in Fig. 6B, this sample contained high quantities of anthraquinones, including purpurin, alizarin, anthragallol and emodin. It suggests that the red dye present in this sample most likely originated from madder (*Rubia tinctorum* L.) or lady's bedstraw (*Galium verum* L.). The presence of brazilin in this sample indicated that brazilwood (*Caesalpinia sappan* L.) was also added to the dyestuff. Sample number 54 contained indigotin and relatively small amount of isatin; these results indicate that the addition of blue dye originated from woad (*Isatis tinctoria* L.). Again, relatively small amount of apigenin was detected. Similarly to the data obtained from the sample number 37, it was most likely a result of the sample contamination by plant material used to prepare the major dye components.

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Sample number 54, as well as sample number 53 (see Table 3) contained not only two red dyes prepared from brazilwood and either madder or lady's bedstraw, but also the addition of indigo. The interpretation of these dyestuffs composition is difficult. Particularly, the addition of blue dye (indigo) is very surprising. The Italian medieval dyer's manual mentions the use of madder and brazilwood combination to obtain the color "*morello*"[60]. It is also known, that madder and brazilwood (without adding blue indigofera or woad indigo) were used to imitate the vermilion red (scarlet) of kermes (*Kermes vermilio*) [23, 24]. Brazilwood used together with indigofera or woad without adding madder allowed to obtain colors like: *"peacock"* (purple shade) [20] or "*paonazzo*" (purple) [21, 22]. Medieval Italian dyer's manual does not mention adding woad or indigofera indigo to madder and brazilwood, therefore it excludes Italian origins of textiles under study. The combination of these three dyes was identified in silk textiles originating from Central Asia [61] and Spain [62]. To confirm this hypothesis further study of the dyeing methods used in Spain and in the Middle East are needed.

Results of the blue samples analysis (including the two objects already discussed in this section) are summarized in Table 3. The relative ratio of isatin to indigotin clearly indicates that the source of blue dye was *Isatis tinctoria*.

As listed in Table 3, natural dyes detected in samples number 37, 86, 88, 91 and 92 were indigo and orchil. The presented samples are interesting examples of combining these two dyes in order to obtain different shades of blue and purple. Depending on the amount of indigo and orchil different shades of blue were obtained: navy blue, bluish-violet or bluish-gray. According to the medieval dyer's manual the silk was dyed first with orchil and then submerged in the indigofera or woad vat [18, 26, 63].

4. Summary and conclusions

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In this work, a novel LC/MS/MS method for identification of indigo, red dyes (anthraquinone, soluble redwoods and orchil) and yellow dyes (flavonoid-based) was developed. The developed method is selective and sensitive, allowing for identification of the listed dyestuff components in the 14th and 15th century silk textiles. For the first time analysis of the major orchil components (orcein derivatives) were conducted with LC/MS/MS. Elemental formulas and retention times of the orcein derivatives were confirmed with HR-MS by carrying out LC-TOF/MS analysis, thereby providing a tool for unambiguous identification of these compounds in the historical samples under study.

Before analyzing historical samples, analysis of the natural indigo were carried out in order to evaluate if isatis to indigotin ratio can be used to elucidate the plant source of indigo detected in historical samples. Commercial *Indigofera* indigo powder showed significantly higher value of this ratio (above 0.5) than *Isatis tinctoria* for which this ratio took values below 0.2. This proportion was used for the identification of blue dye's source.

Results of the presented case-study underline that the developed method can be used to identify the studied natural dyes in medieval silk textiles. Also, identification of natural dyes can be carried out even on very small samples. Furthermore, the developed method allows to understand the technology used in production of material under study.

For the first time the dyes from one of the largest collections of medieval silk textiles from St. Mary's Church in Gdansk, stored in National Museum in Gdańsk, were successfully studied using LC/MS/MS. The obtained data provided valuable insights into the knowledge of manufacturing technology and dyeing techniques of the silk textiles from this collection. The results of this work will also be used for a more precise dating of the studied historical textiles. Using obtained data may allow to identify the origins of studied objects by connecting the dyeing technique to a specific weaving center.

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Figures captions

Figure 1 HR EIC chromatograms of the orcein derivatives

Figure 2 SRM chromatograms of indigotin, indirubin (positive ionization mode) and isatin (negative ionization mode) detected in commercial *Isatis tinctoria* pigment

Figure 3 Relative content of indigotin, indirubin and isatin in natural indigo

Figure 4 Tree diagrams of natural indigo and samples dyed in the laboratory

Figure 5 Picture of the Italian 14th century silk lampas – sample number 37 (A), photograph of the lampas fragment illustrating orchil degradation (B) and the SRM chromatograms of the sample number 37 (C)

Figure 6 Picture of the lampas dated to 14th or 15th century from the orphrey of the chasuble (A) and SRM chromatograms of the sample number 54 (B)

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Table 1 List of the analytes and quantitative method parameters

Compound name	Ions monitored in the SRM mode: Q1/Q3 (m/z)	Retention time (min)	Linear regression coefficient (R^2)	Average recovery (%) *	Limit of detection (µg/ml)	Limit of quantification (µg/ml)
			Negative ionization mode			
Quercetin	301/151, 301/179	19.7	0.999	106 ± 5	0.04	0.1
Apigenin	269/117, 269/151	23.0	0.999	103 ± 4	0.004	0.01
Luteolin	285/149, 285/198	20.7	0.998	105 ± 10	0.1	0.3
Rhamnetin	315/165, 315/300	25.8	0.998	105 ± 10	0.001	0.003
Morin	301/151, 301/163, 301/125	20.6	0.998	107 ± 20	0.01	0.04
Kaempferol	285/151, 285/185	23.6	0.999	104 ± 10	0.01	0.02
Rutin	609/300, 609/271	15.1	0.999	102 ± 4	0.003	0.009
Genistein	269/133	23.0	0.993	109 ± 17	0.02	0.06
Fisetin	285/135, 285/121	18.3	0.999	107 ± 7	0.002	0.004
Purpurin	255/227, 255/183, 255/153	28.9	0.995	106 ± 16	0.005	0.01
Alizarin	239/210, 239,195, 239/210	26.1	0.999	99 ± 3	0.008	0.02
Anthragallol	255/153, 255/227	23.4	0.997	107 ± 5	0.009	0.02
Emodin	269/225, 269/241	32.3	0.999	107 ± 19	0.002	0.005
Hematein	299/174	13.1	0.998	107 ± 13	0.09	0.3
Hematoxylin	301/179	9.4	0.999	102 ± 5	0.01	0.02
Brazilin	285/163	12.3	0.998	113 ± 14	0.06	0.2
Brazilein	283/265	15.8	0.998	98 ± 11	0.2	0.6
Isatin	146/118	13.3	0.998	101 ± 9	0.01	0.03
			Positive ionization mode			
α-amino orcein	363/348	19.5	0.998	105 ± 11	7	18
α –hydroxy orcein	364/349	24.2	0.996	95 ± 13	14	39
β-Amino orcein imine	484/468	17.0/19.5	0.997	100 ± 13	116	314

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γ-Amino orcein imine	484/468	17.0/19.5	0.995	107 ± 21	121	326
β-Amino orcein	485/470	18.0/21.1	0.996	110 ± 24	127	352
γ-Amino orcein	485/470	18.0/21.1	0.994	98 ± 14	105	292
β-Hydroxy orcein	486/469	21.2/24.4	0.999	105 ± 8	61	187
γ-Hydroxy orcein	486/469	21.2/24.4	0.994	109 ± 27	98	297
Indigotin	263/77	29.2	0.997	103 ± 11	0.02	0.07
Indirubin	263/77	30.5	0.999	102 ± 5	0.09	0.3

*Uncertainty is two values of the standard deviation

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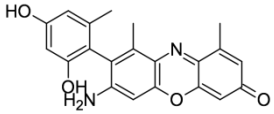
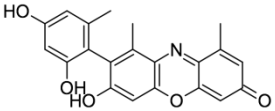
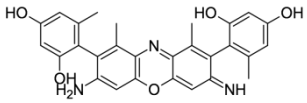
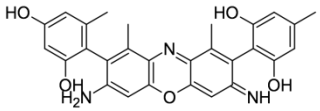
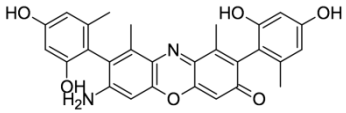
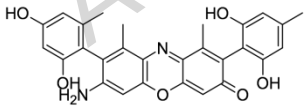
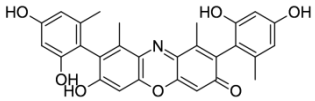
Table 2 Known orcein components

Name	Structure	Elemental composition	m/z measured - $[M+H]^+$	m/z calculated - $[M+H]^+$	Difference (ppm)
α-amino orcein		$C_{21}H_{18}N_2O_4$	363.1340	363.1339	-1.29
α –hydroxy orcein		$C_{21}H_{17}NO_5$	364.1180	364.1179	-0.14
β-Amino orcein imine		$C_{28}H_{25}N_3O_5$	484.1875	484.1867	1.1
γ-Amino orcein imine		$C_{28}H_{25}N_3O_5$	484.1875	484.1867	1.1
β-Amino orcein		$C_{28}H_{24}N_2O_6$	485.1710	485.1707	-0.59
γ-Amino orcein		$C_{28}H_{24}N_2O_6$	485.1710	485.1707	-0.59
β-Hydroxy orcein		$C_{28}H_{23}NO_7$	486.1547	486.1547	0.26

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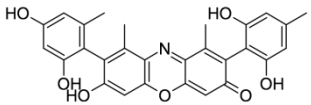
γ-Hydroxy orcein		$C_{28}H_{23}NO_7$	486.1547	486.1547	0.26

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Table 3 Results of the LC/MS/MS analysis for the selected historical textiles

Sample number	Object description	Sample source	Sample color	Sample weight (mg)	Isatin/indigotin ratio	Detected natural colorants*
37	The fragment of chasuble (inv. no. MNG/SD/241/TH) Lampas: Italy, 2nd half of 14th century	Main warp	Sky blue, originally violet	8.16	0.09	Indigotin, α – amino orcein, α – hydroxy orcein, isatin, apigenin
53	The chasuble (inv. no. MNG/SD/242/TH) Lampas: Italy?/Middle East?, 2nd half of 14th century	Main weft	Scarlet	12.77	0.04	Purpurin, alizarin, anthragallol, emodin, apigenin, brazilin, indigotin, isatin
54	The orphrey of the chasuble (inv. no. MNG/SD/250/TH) Lampas: Italy?/Spain?, 14th/15th century	Main warp	Scarlet	6.64	0.08	Purpurin, alizarin, anthragallol, emodin, apigenin, brazilin, indigotin, isatin
86	The pluvial (inv. no. MNG/SD/233/TH) Lampas: Italy?/Middle East?, 14th century	Pattern weft	Violet-blue	5.85	0.001	Indigotin, α – hydroxy orcein, α – amino orcein, isatin, apigenin, genistein
88	The chasuble (inv. no. MNG/SD/267/TH) Lampas: Italy?/Middle East?, 15th century	Main warp	Blue-violet	4.78	0.01	Indigotin, isatin, α – hydroxy orcein, α – amino orcein, apigenin
91	The chasuble (inv. no. MNG/SD/269/TH) Lampas: Italy, 15th century	Main warp	Navy blue	3.36	0.03	Indigotin, α – hydroxy orcein, α – amino orcein, isatin, apigenin, genistein

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92	The sides of dalmatic (inv. no. MNG/SD/276/TH) Lampas: Italy, 2nd half of 14th century	Main warp	Bluish-grey	2.14	0.01	Indigotin, α – hydroxy orcein, α – amino orcein, β – hydroxy orcein, isatin, apigenin

***** Listed in order from highest to lowest concentration

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Fig. 1

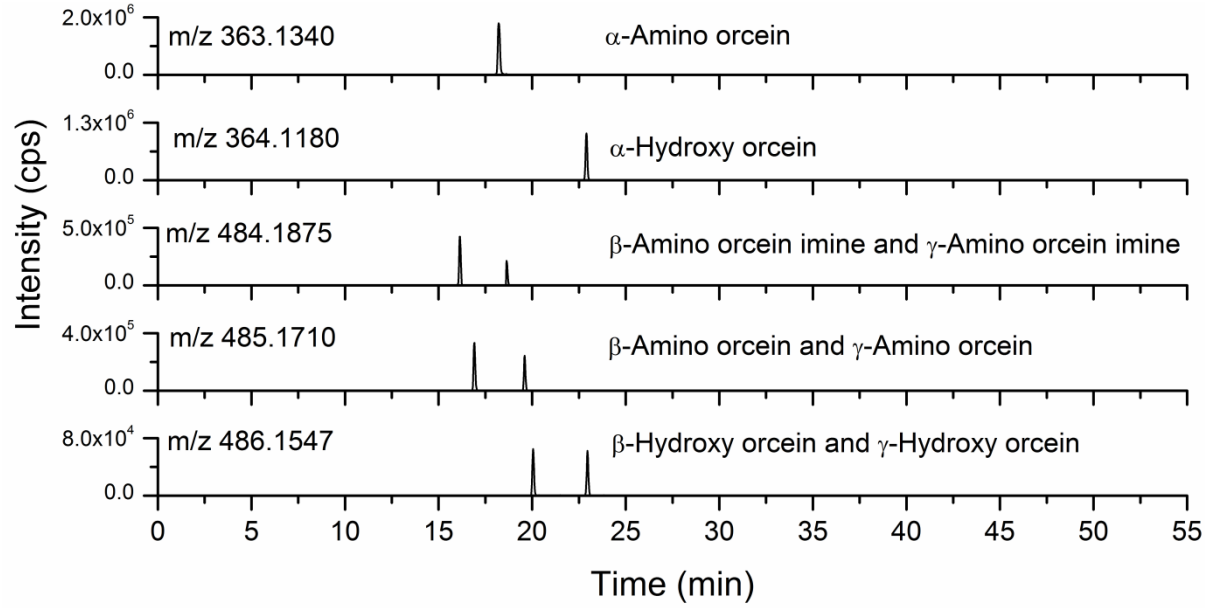

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Fig. 2

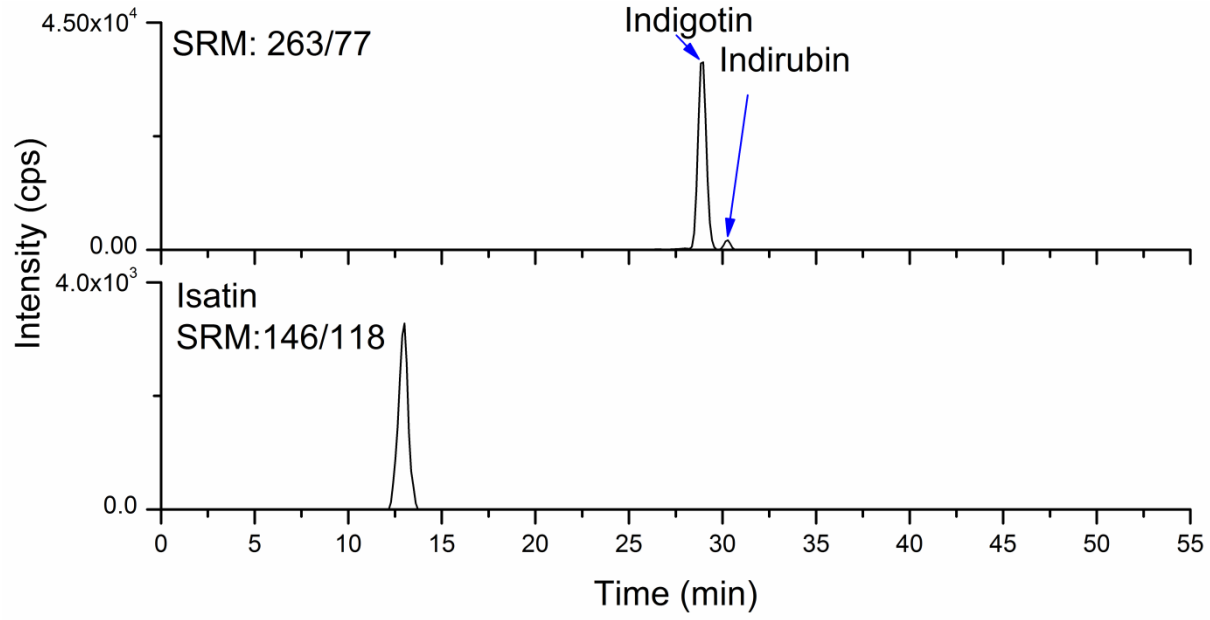

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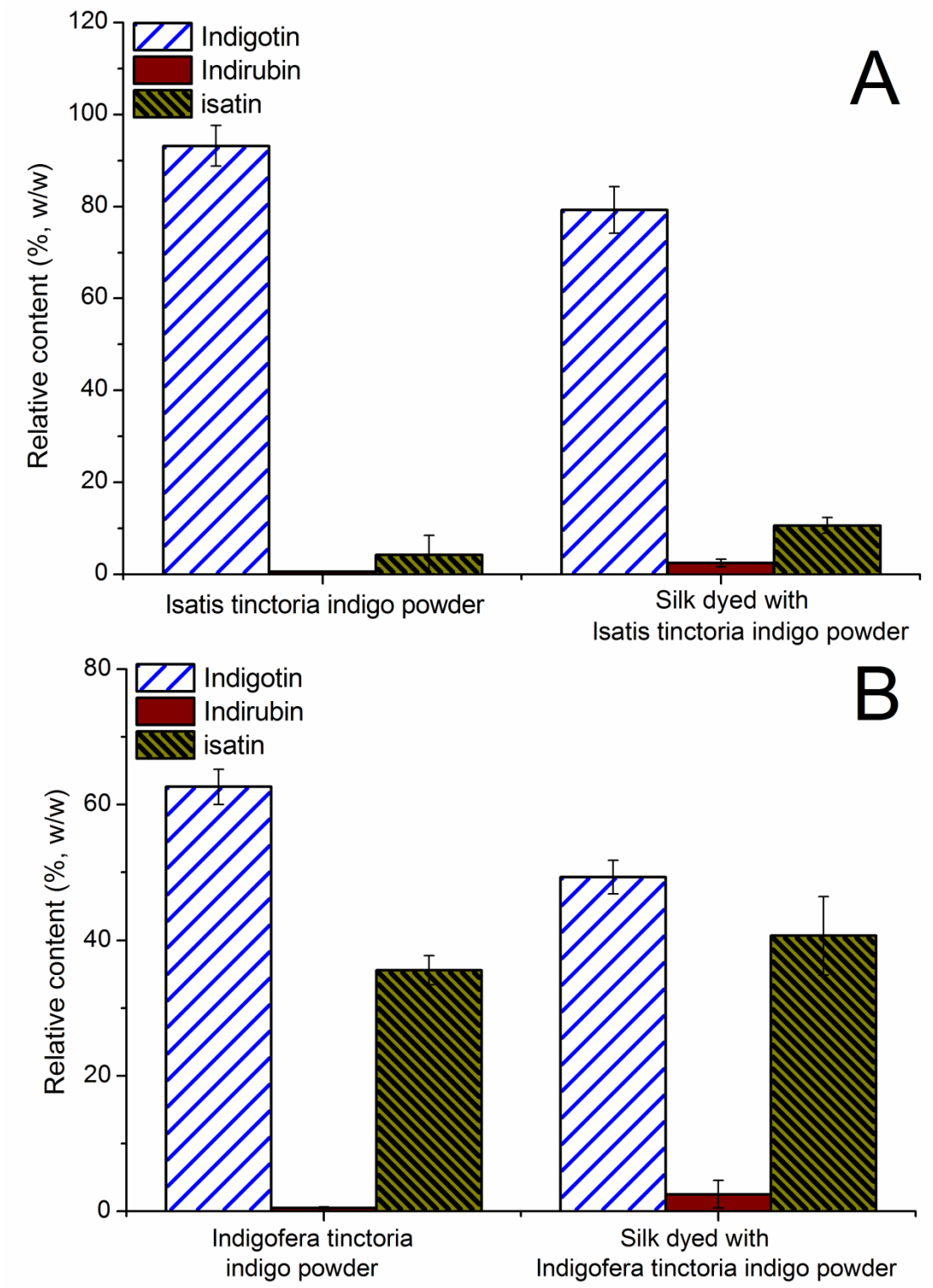
Fig. 3



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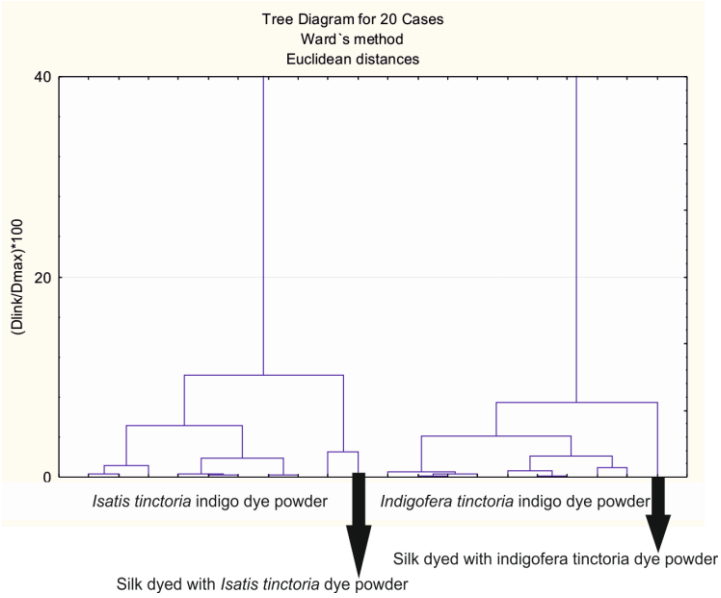

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Fig. 5

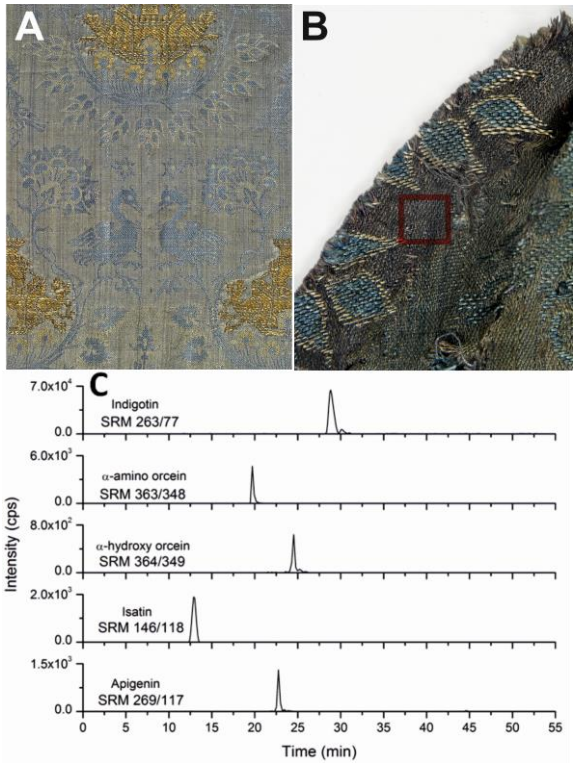

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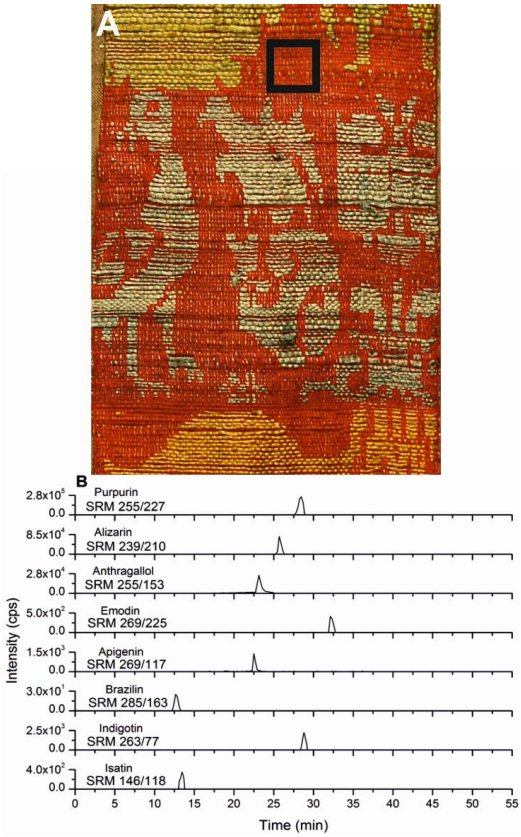
Fig. 6



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Highlights

- Analysis method for orchil and selected natural dyes identification was developed
- Orcein was identified in the medieval textiles with LC/MS/MS
- Natural indigo composition was correlated with the plant source of this dye
- Dyes in the international collection of medieval silk paraments were studied
- Obtained data provided insights into the medieval dyeing techniques and technology

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